Project 2

- Group size: 2
- 15-minute live (English) presentation via Zoom: limited number of PPT slides.
  - I will grade based only on the presentation. I will try to ask many questions.
  - Slots (after quiz 2) will be released in Week 13.
- Two versions
  - Without data: recommended for those who have done classification before.
  - With data: recommended for those who have **not** done classification before.
    - AI could finish Project 2 easily.

Version 1: Without Data (≤3 Slides)

- A supervised learning project motivated by your internship, job, hobby, …
  - Do **not** need to be a classification project.
- Background: What do you want to predict, and why do you want to do it?
  - Potentially many clarification questions.
- Which method do you want to use? Why?
  - Many questions based on the concepts we learned.
- If you've already told me about your project, you could help your teammate or do the version with data.

Version 2: With Data ($\leq$ 6 Slides)

- Classification project
  - The evaluation loss shall deviate from the misclassification loss based on the context.
- Interpretability and performance
  - You must implement at least one glassbox and at least one blackbox model.
- Each model shall have two versions (**train loss**): L2 loss vs. other surrogate losses.
  - You shall separately tune the hyperparameters.
- Compare all of them using **validation AUC** (not train loss) and make a decision.
  - For the chosen model, select the threshold that minimizes **evaluation loss.**

Version 2: With Data

- Are the models calibrated? [Later this class]
  - If not, could you adjust the probability predictions to calibrate them?
- SHAP (Often used in the Capstone project)
  - Even if you choose a glass-box model, you still need to apply tree SHAP to a tree-based blackbox model.
  - The interpretability part must contain both local and global SHAP.

Version 2: Where to Find Dataset?

- https://www.kaggle.com/
  - Find the datasets with high usability ratings!
  - Copying the examples from Kaggle will hurt your grade. (I usually find this out when two teams have extremely similar mistakes/code).
- https://beta.data.gov.sg/datasets
  - Find some Singapore-related datasets!
- Be smart about choosing the dataset: avoid severe imbalance issues/large datasets/data cleaning/missing data…

Common Issues Before

- How are predictions used?
  - You could come up with your own setting. This leads to a customized loss.
- You shall consider tweaking the threshold in label prediction.
  - Tweak it to generate the best customized loss.
- If you obtain almost a perfect prediction, you should consider deleting some features to make the prediction harder.

Project 1 Continued [5 pts]

- Sharpe ratio, mean, and standard deviation for your portfolio using the new data.
- Compare with the corresponding numbers in the validation 2 data.
- Any questions before proceeding?

Last Week: Logistic Regression + SVM in Modern View

- Directly predicting labels using a linear structure
- Different losses: Binomial deviance (logistic) + Hinge (SVM)
  - Binomial deviance and hinge are good surrogate losses for misclassification
  - L2 loss is not a good surrogate loss for misclassification

Logistic Regression + SVM In Historical View

- Logistic regression:
  - Maximum likelihood estimator & predicting the probability of +1 using a link function
- SVM:
  - Maximal margin + soft margin (violation) (+ kernel)

Interpretability

- Glassbox models:
  - Linear structure: LASSO ≫ Ridge
  - Single-tree structure: shallow tree
  - Sum-of-trees structure: EBM (many trees) + FIGS (few trees)
- Blackbox models:
  - We could get some understanding using different tools
  - Local understanding and Global understanding

Interpretability Tools

- Feature Importance: global understanding
- LIME: local understanding
- SHAP: both local and global
- All understandings are based on strong assumptions: be careful

SHAP (Most Popular One)

- Shapley value: calculation + why is it connected to the interpretability of prediction?
- Two Issues:
  - Computational: general approach + tailored approach for tree structure
  - Missing feature: general approach + tailored approach for tree structure
- Any questions before proceeding?
  - Next, we will motivate the overfitting the validation set

If AI Could Win Kaggle, What Is Our Contribution?

- Kaggle competition: given a dataset and an evaluation loss, compete on performance
- Here is a blog regarding using AI to win a Kaggle competition
- Our contribution:
  - Are we asking the right question? Is it a prediction task?
  - Which data to get? How to get data?
  - Which loss to use?
  - Given the dataset and loss, how to trade off interpretability and performance?

Example: Churn Prediction (Used in the Blog)

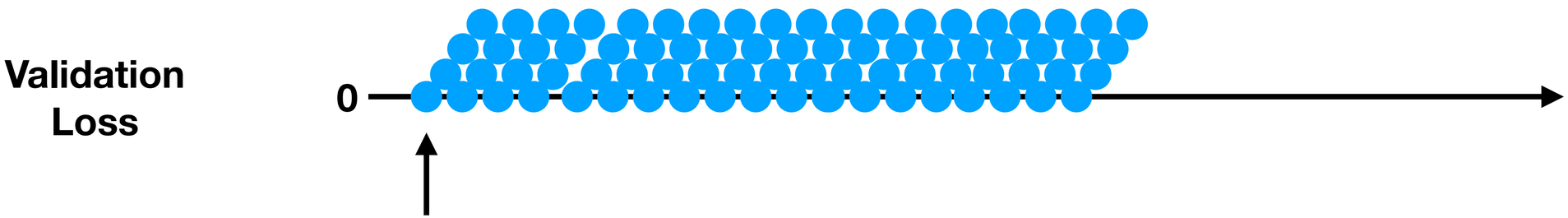
- Instead of prediction (association), should it focus on causality?
- What features are important for the target? Could I get them?
- What are the different losses for different types of misclassification?
- Do I want to use a glassbox model to get interpretability? Or do I want to use a blackbox model but with some interpretability tools?
- In practice, it is very rare to only care about performance.
- Next, we will talk about common issues in getting good performance.
    - Have good habits and avoid using AI to overfit the validation set

Common Issues in Getting Good Performance

- Before AI: Implementations of explorations are hard
- With AI: Implementations of explorations are easy
  - Without a clear understanding
  - With a clear understanding
- Recommendation

AI without a Clear Understanding

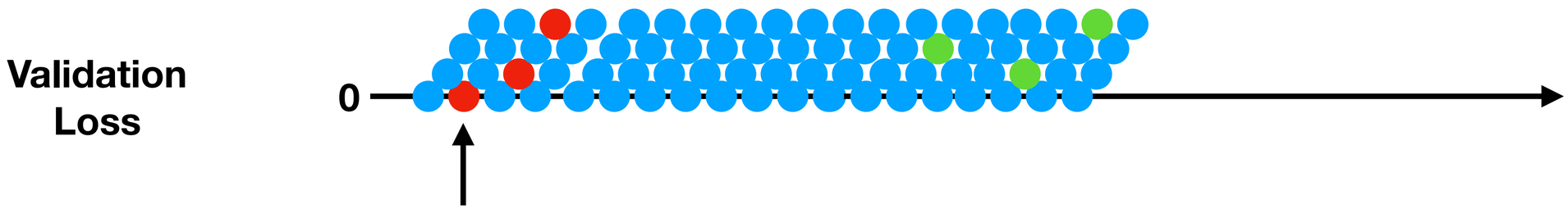
Too many models are tried, do not know which one to choose



≪ Test Loss: severe optimizer's curse because too many models were tried.

AI with a Clear Understanding [Easy]

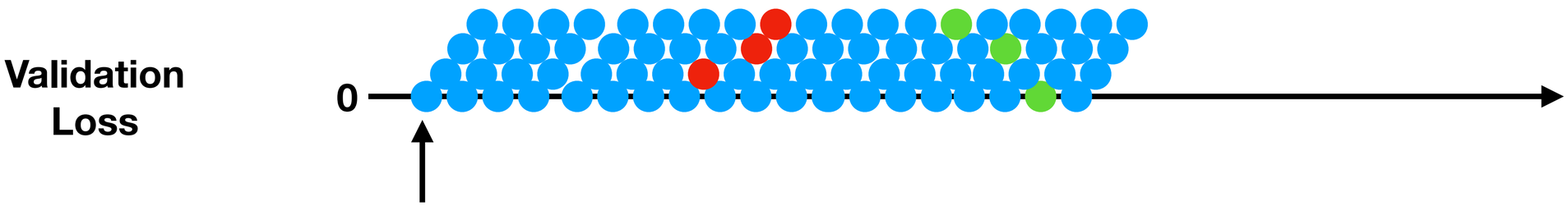
Few Red explorations work well. Choose the best Red instead of blue.



≈ Test Loss: no severe optimizer's curse because few reds are tried

AI with a Clear Understanding [Hard]

Dilemma: best blue v.s. best red

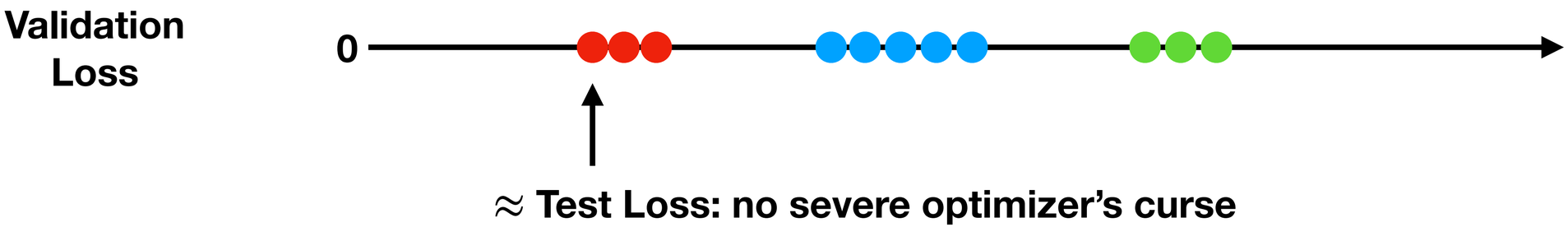


≪ Test Loss: optimizer's curse because too many blues are tried

Few Red explorations do not work well.

AI with a Clear Understanding: Recommended Practice

Few reasonable explorations



Do not try too hard!

Next, overfitting of the validation set